A history criterion for the non-axisymmetric part of a viscous flow

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Abstract

We give a sufficient regularity criterion when the angular mean of a three-dimensional Navier–Stokes velocity satisfies prescribed anisotropic bounds. The remaining assumptions are two time integrals of the angular fluctuation. A radial circulation barrier preserves a spatial factor in the Reynolds torque, while a divergence representation of the Reynolds stress avoids differentiating the fluctuation in the ancient-limit argument. For finite-energy whole-space flows, one weighted amplitude integral suffices: the exact defect equation supplies the missing gradient history on short intervals following the mean flow, with the pressure tail retained. The proof adapts the axisymmetric argument of Constantin, Ignatova, and Vicol. This is an analytic conditional criterion, not a Lean theorem or an arbitrary-data regularity proof.

1 Statement

Let $Q = B(1) \times (-1, 0)$ and let (u, p) be a suitable weak solution of

$$u_t + (u \cdot \nabla)u - \Delta u + \nabla p = 0, \qquad \operatorname{div} u = 0,$$

smooth on compact subsets of Q. Assume the usual local energy class $u \in L^\infty_t L^2_x \cap L^2_t H^1_x$ and $p \in L^{3/2}(Q)$. For rotations Q_ϕ about the fixed third axis put

$$\mathcal{P}u(x) = \frac{1}{2\pi} \int_0^{2\pi} Q_\phi u(Q_\phi^{-1} x)\, \mathrm{d}\phi, \qquad v = \mathcal{P}u, \qquad w = u - v.$$

The same notation denotes the induced action on tensors. Write $r = |x'|$, $z = x_3$, and $\tau = -t$. Unless a smaller ball is specified, suprema are on $B(1)$.

Theorem 1. *Suppose $0 < h < 1/2$ and, for every $a + b \le 2$,*

$$|\partial_r^a \partial_z^b v_r| \le C \tau^{-1/2 - a/2 - (1/2-h)b}, \tag{1}$$

$$|\partial_r^a \partial_z^b v_z| + |\partial_r^a \partial_z^b v_\theta| \le C \tau^{-1/2 - h - a/2 - (1/2-h)b}. \tag{2}$$

Set $K(t) = \|w(t)\|_\infty$ and $\mathcal{T}(t) = \|(\mathcal{P}[(w \cdot \nabla)w])_\theta\|_\infty$. If

$$\int_{-1}^0 K(t)^2 \, \mathrm{d}t < \infty, \qquad \int_{-1}^0 \sqrt{-t}\, \mathcal{T}(t) \, \mathrm{d}t < \infty, \tag{3}$$

then $(0, 0)$ is a regular point of u.

Angular averaging in $\mathcal{T}$ precedes the absolute value. The mean bounds are those used in [1]. Their validity and (3) are hypotheses, not consequences of the energy inequality. The axis and the ball are fixed; the theorem makes no assertion about propagating approximate symmetry from a shrinking core.

Corollary 2. *In Theorem 1, a sufficient replacement for (3), with $L(t) = \|\nabla w(t)\|_\infty$ using Cartesian derivatives, is*

$$K(t) \le C\tau^{-\beta}, \qquad L(t) \le C\tau^{-\beta-\sigma}, \qquad \beta < \tfrac{1}{2}, \quad 2\beta + \sigma < \tfrac{3}{2},$$

with $\beta, \sigma \ge 0$. In particular, the derivative cost $\sigma = 1/2$ permits every $\beta < 1/2$.

Indeed $\mathcal{T} \le KL$, so a sufficient history condition is $\int K^2 < \infty$ and $\int \sqrt{-t}\, KL < \infty$. These are two separate bounds. No derivative estimate is inferred from the amplitude estimate in this corollary. Section 6 gives such an implication under additional whole-space hypotheses. Another sufficient choice near zero is $K \le C\tau^{-1/2}\log(e/\tau)^{-p}$ and $L \le C\tau^{-1}\log(e/\tau)^{-q}$, where $p > 1/2$ and $p + q > 1$.

2 Two scalar history estimates

Lemma 3. *Let $f \ge 0$ be continuous on compact subsets of $(0, a]$, with $J_f = \int_0^a \sqrt{s} f(s)\,\mathrm{d}s < \infty$, and define $B(\tau) = \int_\tau^a f(s)\,\mathrm{d}s$. Then*

$$\sqrt{\tau} B(\tau) \longrightarrow 0, \qquad \int_0^a s^{-1/2} B(s)\,\mathrm{d}s = 2J_f. \tag{4}$$

For $0 \le \eta < 1/2$,

$$\left\| \tau^{-\eta} \int_\tau^a s^\eta f(s)\,\mathrm{d}s \right\|_{L^2(0,a)} \le \frac{J_f}{\sqrt{1-2\eta}}. \tag{5}$$

Proof. For the limit, apply dominated convergence to $\mathbf{1}_{s\ge\tau}\sqrt{\tau/s}\,\sqrt{s} f(s)$. The equality is Tonelli's theorem. Minkowski's integral inequality gives

$$\left\| \int_0^a \mathbf{1}_{\tau<s} \tau^{-\eta} s^\eta f(s)\,\mathrm{d}s \right\|_2 \le \int_0^a s^\eta f(s) \left(\int_0^s \tau^{-2\eta}\,\mathrm{d}\tau \right)^{1/2} \mathrm{d}s,$$

which is (5). Truncation justifies the calculation before passing to the possibly singular endpoint. □

Lemma 4. *Suppose an axisymmetric circulation $\Gamma = r v_\theta$ satisfies*

$$\Gamma_t + v_r \Gamma_r + v_z \Gamma_z = (\partial_{rr} - r^{-1}\partial_r + \partial_{zz})\Gamma + r F_\theta,$$

on a ball for $-a < t < 0$, with bounded initial and lateral values, $\Gamma = 0$ on the axis, $|v_r| \le C_r(-t)^{-1/2}$, and $|F_\theta| \le f(-t)$ as in Lemma 3. Then

$$|\Gamma(r,z,t)| \le A(-t) + rB(-t), \quad A(\tau) = C_0 + C_r \int_\tau^a s^{-1/2} B(s)\,\mathrm{d}s, \tag{6}$$

where C_0 bounds the initial and lateral values. In particular A is bounded, and $\sqrt{\tau}B(\tau) \to 0$.

Proof. For $r > 0$, applying the circulation operator to $\Phi(r,t) = A(-t) + rB(-t)$ gives

$$C_r(-t)^{-1/2} B + rf + v_r B + B/r \ge rf.$$

Thus Φ dominates both Γ and $-\Gamma$ by the maximum principle on each compact time prefix. A positive maximum of their difference cannot occur on the axis, where $\Gamma = 0$ and $\Phi \ge 0$; no differentiability of r at the axis is required. The bounds follow from Lemma 3. □

3 The mean equation and its rescaled stress

Rotation covariance and $\mathcal{P}w = 0$ give the exact equation

$$v_t + (v\cdot\nabla)v - \Delta v + \nabla\mathcal{P}p = F, \qquad F = -\mathcal{P}[(w\cdot\nabla)w] = -\operatorname{div}\mathcal{P}(w\otimes w). \tag{7}$$

In particular $\|F(t)\|_\infty \le K(t)L(t)$, using the Euclidean operator norm for ∇w. Other tensor norm conventions change only a fixed constant.

The original solution has a regular spherical annulus inside any prescribed interval of radii. Indeed, partial regularity makes the top-slice singular set have zero $\mathcal{H}^1$ measure. Its image under $x \mapsto |x|$ has measure zero, so some intervening sphere is regular. Compactness and interior estimates give an annulus on which u, ∇u, and $\nabla^2 u$ are bounded through the endpoint, as in the first part of [1, Lemma 3.2]. Its rotational invariance gives the same bounds for v and w. This argument uses suitability of u; it does not require a partial-regularity theorem for (7).

Choose a sphere in this annulus, and take $f(\tau)$ to be the supremum of $|F_\theta(-\tau)|$ on its enclosed closed ball. This is continuous on compact time prefixes and satisfies $J_f < \infty$ by (3). Lemma 4 applies to the mean.

Write the meridional entries of $\mathcal{P}(w\otimes w)$ in the cylindrical frame as $a = R_{rr}$, $b = R_{\theta\theta}$, $c = R_{zz}$, and $d = R_{rz}$. Their absolute values are at most K^2. Direct differentiation yields

$$F_r = -a_r - (a-b)/r - d_z, \qquad F_z = -d_r - d/r - c_z, \tag{8}$$

$$\frac{(\operatorname{curl} F)_\theta}{r} = \left(\partial_{rr} + \frac{3}{r}\partial_r - \partial_{zz}\right)\frac{d}{r} + \frac{1}{r}\partial_{rz}(c-a) - \frac{1}{r^2}\partial_z(a-b). \tag{9}$$

The quantities here are smooth at the axis in the appropriate lifted variables. For a distributional estimate it suffices that $d = O(r)$.

Consider the rescaling used in [1, Section 4]: for $t_n \uparrow 0$,

$$\tau_n = -t_n, \quad \lambda_n = \tau_n^{1/2}, \quad \delta_n = \tau_n^h, \quad \mu_n = \lambda_n/\delta_n, \quad r = \lambda_n R, \quad z = z_n + \mu_n Z, \quad t = \tau_n s.$$

The mean components are normalized as $V_n = \lambda_n v_r$, $W_n = \lambda_n\delta_n v_z$, $S_n = \lambda_n\delta_n v_\theta$, and the potential vorticity as $\Omega_n = \lambda_n^3\delta_n(\operatorname{curl} v)_\theta/r$. For a stress component a, let a_n denote its composition with this change of variables, without an amplitude normalization.

Lift $R = |X|$ to four radial dimensions. Since

$$\frac{1}{R}\partial_R H = \operatorname{div}_X\left(\frac{XH}{R^2}\right) - \frac{2H}{R^2},$$

the rescaled forcing in the potential-vorticity equation is

$$\begin{aligned} G_n = \lambda_n^2\delta_n\Big[&(\Delta_X - \delta_n^2\partial_{ZZ})(d_n/R) \\ &+ \delta_n\partial_Z\left\{\operatorname{div}_X\frac{X(c_n-a_n)}{R^2} - \frac{2(c_n-a_n)+(a_n-b_n)}{R^2}\right\}\Big]. \end{aligned} \tag{10}$$

Both R^{-1} and R^{-2} are locally integrable in $\mathbb{R}^4$. The divergence identity has no axis-supported mass: the flux through $R = \epsilon$ is $O(\epsilon^2)$ for bounded H. Move all derivatives in (10) onto a compactly supported smooth test function with time support $J \Subset (-\infty, 0)$. The resulting absolute value is bounded by

$$C_J\lambda_n^2\delta_n\int_J K(\tau_n s)^2\,\mathrm{d}s = C_J\delta_n\int_{\tau_n J} K(t)^2\,\mathrm{d}t \longrightarrow 0. \tag{11}$$

Thus no derivative of the fluctuation is needed in this part of the limiting argument. In a recentered zoom with $r = \lambda_n(A_n + R)$ and $A_n \to \infty$, the azimuthal-vorticity force has the same prefactor $\lambda_n^2\delta_n$ multiplying at most two derivatives of the stress; its additional coefficients $(A_n + R)^{-1}$ and $(A_n + R)^{-2}$ are uniformly bounded on compact sets. Estimate (11) therefore applies there too.

4 Meridional smallness

Define, on a smaller ball,

$$G_v(t) = \big\| |\partial_r v_r| + |v_r/r| + |\partial_z v_z| + |\partial_z v_r| \big\|_\infty .$$

We verify the two changes needed in the proof of [1, Proposition 1.5] to conclude

$$(-t)G_v(t) \longrightarrow 0. \tag{12}$$

First, the circulation barrier gives, in the finite-axis case,

$$|RS_n| \le \delta_n A(\tau_n|s|) + \delta_n\lambda_n RB(\tau_n|s|) \longrightarrow 0$$

uniformly on compact rescaled time slabs and bounded R. The mean bounds also give $|S_n/R| \le C$ on those slabs, including the axis. Splitting at $R = \epsilon$ shows that S_n^2/R^2 tends to zero locally in L^1 in the lifted space. In the receding-axis case use instead

$$|S_n| \le \frac{\delta_n A(\tau_n|s|)}{A_n + R} + \delta_n\lambda_n B(\tau_n|s|).$$

Second, the force is small in a negative Sobolev norm integrated in time, rather than uniformly small as a function. Away from the axis, (10) and (11) give

$$\|G_n\|_{L^1(J;W^{-2,p}(D))} \longrightarrow 0$$

for a fixed rectangle D and any finite p. The analogous assertion holds in the receding-axis case. The other terms in the rescaled vorticity equation are uniformly bounded in this negative norm. Together with the uniform spatial Lipschitz bound supplied by (1)–(2), this gives an increment bound $C|s - s'| + \epsilon_n$ with $\epsilon_n \to 0$. Interpolation gives asymptotic uniform time equicontinuity, also at the selected upper time. Finitely many initial members are smooth, so Arzelà–Ascoli still applies. Boundedness and the small volume of $R < \epsilon$ give convergence across the axis.

The remaining ancient-limit argument is unchanged: the force and swirl source vanish, and the mean bounds supply the backward decay and bounded Lipschitz drift required by [1, Lemma 3.3]. The limiting vorticity vanishes. At the selected time, incompressibility and boundedness of the limiting meridional velocity then contradict a nonzero selected derivative, in either axis case. This proves (12). No bound on $\partial_r v_z$ is asserted.

5 Full-velocity enstrophy

Choose a radial cutoff χ whose derivative is supported in a regular annulus, and write

$$Y = \|\chi \operatorname{curl} u\|_2^2 , \quad M = \|\chi\nabla \operatorname{curl} u\|_2^2 , \quad V_\theta = \|v_\theta\|_\infty .$$

Here V_θ is taken on a ball containing the support of χ, with its boundary in the regular annulus; K is still taken on $B(1)$. All cutoff errors are uniformly bounded. Weighted L^2 contraction of

angular averaging, including its tensor action on gradients, gives $Y_v \le Y$ and $M_v \le M$. Local div–curl estimates consequently bound the first derivatives of u, v by $C(Y+1)^{1/2}$ and their second derivatives by $C(M+1)^{1/2}$ in these weighted norms.

For $\Pi_\chi(q) = \int \chi^2 (\operatorname{curl} q \cdot \nabla) q \cdot \operatorname{curl} q$, put $\omega_q = \operatorname{curl} q$. There is an exact telescoping identity

$$\begin{aligned}\Pi_\chi(u) - \Pi_\chi(v) = \int \chi^2 \big[&(\omega_w \cdot \nabla) u \cdot \omega_u + (\omega_v \cdot \nabla) w \cdot \omega_u \\ &+ (\omega_v \cdot \nabla) v \cdot \omega_w\big].\end{aligned} \tag{13}$$

Integrate the derivative on w off that factor in each term. The remaining factors have derivative orders one and two, apart from bounded cutoff errors. Hence, for every $\epsilon > 0$,

$$\begin{aligned}|\Pi_\chi(u) - \Pi_\chi(v)| &\le CK(Y+1)^{1/2}(M+1)^{1/2} + C \\ &\le \epsilon M + C_\epsilon (K^2+1)(Y+1).\end{aligned}$$

For the mean's stretching, the cylindrical calculation in [1, Section 5] gives

$$\Pi_\chi(v) \le \epsilon M + C_\epsilon (G_v + V_\theta^2 + 1)(Y+1).$$

In particular its large $\partial_r v_z$ term is integrated by parts using $(\omega_v)_r = -\partial_z v_\theta$, with v_θ left undifferentiated. Testing the original unforced vorticity equation and absorbing both small dissipation terms gives

$$Y' + cM \le C_*[G_v + V_\theta^2 + K^2 + 1](Y+1). \tag{14}$$

Fix $0 < \eta < 1/2$ and shorten the final interval so that $|v_r/r| \le \eta/\tau$, using (12). The equation for $\tau^\eta v_\theta$ has nonnegative zeroth-order coefficient. Its maximum principle therefore yields

$$V_\theta(-\tau) \le C_\eta \tau^{-\eta} + \tau^{-\eta} \int_\tau^a s^\eta f(s)\, \mathrm{d}s.$$

Lemma 3 proves $V_\theta \in L^2_t$. Since $K \in L^2_t$ as well, Gronwall in (14) gives

$$Y(t) + 1 \le C_\eta (-t)^{-C_* \eta}.$$

Here C_* was fixed before η. Choose η with $2C_*\eta < 1$. Local Sobolev now places u in $L^4_t L^6_x$ on a smaller cylinder. The local criterion of [2, Theorem 1.1(i)], applied on backward cylinders ending at zero, makes $(0,0)$ regular. This finishes the proof of Theorem 1.

6 Paying the gradient history from the defect equation

We now use the actual evolution of the defect to replace both histories by one amplitude integral. The following whole-space assumptions let us control the pressure produced by localization.

Theorem 5. *Suppose the solution in Theorem 1 is the restriction of a smooth solution on $\mathbb{R}^3 \times (-1,0)$ with $u \in L^\infty((-1,0); L^2(\mathbb{R}^3))$ and canonical pressure. Retain the local suitability assumptions and the mean bounds (1)–(2). It suffices to assume, for some $0 < a < 1$,*

$$\int_0^a \tau^{-h/2} \sqrt{\log(e/\tau)}\, \|w(-\tau)\|^2_{L^\infty(B(1))}\, \mathrm{d}\tau < \infty. \tag{15}$$

In particular, the amplitude estimate $\|w(-\tau)\|_\infty \le C\tau^{-\beta}$ suffices if $\beta < 1/2 - h/4$, without a separate derivative estimate. At equality, the additional factor $\log(e/\tau)^{-p}$ suffices for $p > 3/4$.

The theorem does not assert (15) from initial data. Its proof has a whole-space smoothing estimate and a localization step. Write $\mathbb{L}$ for the Leray projection, to distinguish it from angular averaging $\mathcal{P}$.

Lemma 6. *Let a smooth divergence-free U on $\mathbb{R}^3 \times [0,d]$ satisfy*

$$U_t - \Delta U = -\mathbb{L}\,\mathrm{div}(U \otimes U) + g, \qquad V = \mathcal{P}U, \quad W = U - V, \quad G = (1-\mathcal{P})g.$$

Assume the fields and the spatial derivatives used below are bounded on compact time prefixes, and that $\|V\|_\infty \le M$, $\|\nabla V\|_\infty \le N$, $\|G\|_\infty \le G_0$ there, with $M, N > 0$. Put $K(q) = \|W(q)\|_\infty$, $L(q) = \|\nabla W(q)\|_\infty$, and

$$Z_d = \sup_{0<q\le d} \{K(q) + \sqrt{q}L(q)\}, \qquad \log_+ x = \max(\log x, 0).$$

There are universal constants such that

$$Z_d \le C_0(K(0) + dG_0) + C_1 N d \left[1 + \log_+ \frac{M}{N\sqrt{d}}\right] Z_d + C_2\sqrt{d} Z_d^2. \tag{16}$$

Consequently $Z_d \le 2C_0(K(0) + dG_0)$ if the linear coefficient is at most $1/4$ and $\sqrt{d}(K(0) + dG_0)$ is sufficiently small.

Proof. Rotation covariance gives the exact homogeneous-in-W equation

$$W_t - \Delta W = -\mathbb{L}\,\mathrm{div}\left[V \otimes W + W \otimes V + (1-\mathcal{P})(W \otimes W)\right] + G. \tag{17}$$

In particular the complete defect self-interaction is retained. For the heat kernel H_s, the kernel of $e^{s\Delta}\mathbb{L}\,\mathrm{div}$ is

$$(B_s)_{ijk} = \delta_{ij}\partial_k H_s + \int_s^\infty \partial_{ijk} H_b \,\mathrm{d}b.$$

Differentiating the Gaussian and integrating in b gives

$$|B_s(y)| \le C(\sqrt{s} + |y|)^{-4}, \qquad |\nabla B_s(y)| \le C(\sqrt{s} + |y|)^{-5},$$
$$\|B_s\|_1 \le Cs^{-1/2}, \qquad \int |y|\,|\nabla B_s(y)|\,\mathrm{d}y \le Cs^{-1/2}.$$

The first absolute moment of B_s is not integrable at infinity. Splitting the radial integral at $\sqrt{s}$ and M/N instead gives

$$\int |B_s(y)| \min(N|y|, 2M)\,\mathrm{d}y \le CN\left[1 + \log_+ \frac{M}{N\sqrt{s}}\right], \tag{18}$$
$$\int |\nabla B_s(y)| \min(N|y|, 2M)\,\mathrm{d}y \le CNs^{-1/2}. \tag{19}$$

For each endpoint (x,q) follow the backward characteristic $z'(s) = V(z(s),s)$, $z(q) = x$, and set $a(s) = V(z(s),s)$. Move $a(s)\cdot\nabla W$ to the left side of (17); its two linear tensors now contain $V - a$ instead of V, since $\mathrm{div}\, W = 0$. This identity holds everywhere with this one spatially constant $a(s)$. The translated Duhamel kernels at the chosen endpoint are centered at $z(s)$, where

$$|V(y,s) - a(s)| \le \min(N|y - z(s)|, 2M).$$

For the gradient estimate, differentiate that representation with $a(s)$ held fixed. No derivative of the selected characteristic is introduced: the same representation is valid at every nearby endpoint with this fixed a. Estimates (18)–(19) bound the linear amplitude and gradient terms.

For the quadratic gradient term move the derivative onto the tensor: $\|\nabla(1-\mathcal{P})(W\otimes W)\|_\infty \le CKL$. Angular averaging and its tensor action are contractions in the Euclidean sup norms. The time integral $\int_0^q (q-s)^{-1/2}s^{-1/2}\,\mathrm{d}s = \pi$ then bounds this term by CZ_d^2, before multiplication by $\sqrt{q}$. The quadratic amplitude term is bounded by $C\sqrt{q}Z_d^2$. The initial term and the divergence-free source use the ordinary translated heat kernel. Finally

$$\int_0^q \left[1+\log_+\frac{M}{N\sqrt{s}}\right]\mathrm{d}s \le Cq\left[1+\log_+\frac{M}{N\sqrt{q}}\right].$$

This proves (16). The estimate holds on every shorter prefix, whose linear coefficient is no larger. A continuity bootstrap gives the final claim; smoothness supplies the initial continuity. If $K(0)+dG_0=0$, use uniqueness or a positive bootstrap bound tending to zero. □

Proof of Theorem 5. First suppose a globally bounded defect equation as in Lemma 6 has been obtained, with G uniformly bounded and

$$\|V(-\tau)\|_\infty \le C_M\tau^{-1/2-h}, \qquad \|\nabla V(-\tau)\|_\infty \le C_N\tau^{-1-h}. \tag{20}$$

Let $\ell(\tau)=\log(e/\tau)$ and choose

$$d(\tau)=\frac{c\tau^{1+h}}{\ell(\tau)}, \qquad 0<c<1. \tag{21}$$

On $[-\tau-d,-\tau]$, the linear coefficient in (16) is at most $Cc(1+|\log c|+h)$: here $Nd\le C_Nc/\ell$ and $M/(N\sqrt{d})$ is bounded by a constant times $c^{-1/2}\tau^{-h/2}\sqrt{\ell}$. Take c small enough in terms of the mean bounds.

Write $K=\|W\|_\infty$, $L=\|\nabla W\|_\infty$, and

$$E(\tau)=\int_\tau^{\tau+d(\tau)} K(-s)^2\,\mathrm{d}s.$$

Condition (15), with the localized global norm, implies $K\in L^2_t$ and $E(\tau)\to 0$. There is a time $s_0\in[-\tau-d,-\tau-d/2]$ at which $K(s_0)^2\le 2E(\tau)/d$. Applying Lemma 6 on $[s_0,-\tau]$ is therefore legitimate for all sufficiently small τ; its small parameter is at most $C\sqrt{E(\tau)}+d^{3/2}G_0\to 0$. We obtain

$$K(-\tau)+\sqrt{d}L(-\tau)\le C[d^{-1/2}\sqrt{E(\tau)}+dG_0]. \tag{22}$$

Consequently

$$\sqrt{\tau}K(-\tau)L(-\tau)\le C\sqrt{\tau}d^{-3/2}E(\tau)+C\sqrt{\tau}d^{3/2}G_0^2. \tag{23}$$

The second term is integrable. For the first, Tonelli gives

$$\int_0^{a/2}\sqrt{\tau}d(\tau)^{-3/2}E(\tau)\,\mathrm{d}\tau \le C\int_0^a s^{-h/2}\sqrt{\ell(s)}K(-s)^2\,\mathrm{d}s. \tag{24}$$

Indeed, $\tau\le s\le\tau+d(\tau)$ implies $s/(1+c)\le\tau\le s$. Since d increases, these τ lie in an interval of length at most $d(s)$, while the integrand's coefficient is at most $Cs^{-1-3h/2}\ell(s)^{3/2}$. This proves (24), hence the two histories on a final interval, using $\mathcal{T}\le KL$.

It remains to construct the localization used above. Choose a radial cutoff χ, equal to one on an interior ball and with its transition inside a regular annulus of the original solution. Put

$$\Psi=(-\Delta)^{-1}\operatorname{curl}u, \qquad U=\operatorname{curl}(\chi\Psi)=\chi u+b, \qquad b=\nabla\chi\times\Psi.$$

The L^2 hypothesis and incompressibility give $\operatorname{curl}\Psi = u$; the potential is well defined by its order-minus-one kernel. On the annulus, its separated kernel and every required derivative are bounded against $\|u\|_2$, while its local part is controlled by interior spatial regularity. For canonical pressure, the same decomposition for its order-zero transform of $u \otimes u$ bounds p and its spatial derivatives on the annulus. Likewise

$$\Psi_t = -\operatorname{curl} u - (-\Delta)^{-1} \operatorname{curl}\operatorname{div}(u \otimes u)$$

has bounded spatial derivatives there: its bilinear operator is of order zero, with separated kernels controlled by $\|u\|_2^2$. These statements use only annular regularity and the uniform energy bound, not regularity of the interior through time zero.

With pressure χp, the exact force of U is

$$\begin{aligned} f_U = b_t - 2\nabla\chi \cdot \nabla u - (\Delta\chi)u - \Delta b + p\nabla\chi \\ + (U \cdot \nabla)U - \chi(u \cdot \nabla)u. \end{aligned} \tag{25}$$

It is supported in the regular annulus and uniformly bounded in spatial C^1. Its Leray projection is uniformly bounded, as follows by subtracting $f_U(x)$ in the local principal value and using its L^1 norm for the far part. Thus the equation for U has $g = \mathbb{L}f_U$ and bounded $G = (1-\mathcal{P})g$, including pressure.

Rotations preserve the interior ball, so $U = u$, $\mathcal{P}U = v$, and $U - \mathcal{P}U = w$ there. The transition region contributes bounded fields and derivatives; outside the cutoff $U = 0$. The mean assumptions therefore imply (20) globally. The Cartesian gradient terms v_r/r and v_θ/r are controlled by radial derivatives along the segment to the axis. The global weighted amplitude integral is bounded by (15) plus the finite contribution of a bounded function. Equations (22)–(24) now supply the required histories on a smaller cylinder of the original flow. Theorem 1 applies there. The power and logarithmic consequences follow by integrating $\tau^{-h/2-2\beta}\ell(\tau)^{1/2-2p}$. □

7 Scope of the calculation

The theorem allows an unbounded angular fluctuation, but requires its history to satisfy (3) and separately assumes the mean geometry. Neither hypothesis is established here for general data. Theorem 5 removes a separate gradient hypothesis in the whole-space setting, but its weighted amplitude condition remains unproved for arbitrary data. Its localization constants may depend on the solution on a regular annulus; they are not uniform data-only bounds. An original force uniformly bounded in spatial C^2 can also be included: its torque has finite J_f, its rescaled vorticity source vanishes, and its enstrophy contribution is bounded by $C(Y+1)$. Thus a distinction between forced and unforced flows must enter a proof of the hypotheses, not be inferred from this criterion.

The accompanying symbolic calculation checks the mean equation, cylindrical stress divergence, lifted force identity, and the stretching integration by parts on explicit polynomial fields with all six stress components nonzero. It also checks the Hardy constant on a symbolic power profile. Such calculations check algebra only; the fields are not finite-energy Navier–Stokes solutions, and the program does not certify the analytic proof. No statement in this note has been added as an axiom or a theorem to the Lean library.

References

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